

Robert Charles

Lead Enterprise Architect

508 Vann Rd, Highlands, TX 77562 · 2547089501 · worlddev8081@gmail.com

Enterprise Architect with 9 years of experience in designing and implementing scalable, resilient systems using technologies like AWS, Kubernetes, and Docker. Expert in AI-driven architecture and data model design, executing risk mitigation strategies and maintaining robust IT security architectures. Proven track record in mentoring technical teams, establishing CI/CD processes, and managing solution architecture for cross-functional initiatives. Skilled in translating complex business requirements into actionable architecture frameworks, delivering measurable improvements in system performance and stakeholder alignment.

WORK EXPERIENCE

Lead Enterprise Architect

August 2024 – January 2026

Corelight, Remote

- Architected a microservices ecosystem with Kubernetes and AWS, managing 60+ services to handle 2 million requests per minute.
- Created integration frameworks using RESTful APIs, enhancing cross-system communication and reducing manual data transfers from 4 hours to 15 minutes.
- Directed a shift to AI-first architectures, resulting in a increase in automated security threat detection efficiency.
- Established a central log management system using Elasticsearch and Logstash, facilitating real-time monitoring with over 10 TB of data indexed daily.
- Implemented security architecture principles ensuring zero major breaches across 100 distinct deployments.
- Standardized CI/CD using Jenkins and AWS CodePipeline, slashing deployment times from 2 hours to 15 minutes.
- Engineered data migration tools for transitioning 50 TB of legacy data with AWS Glue and PySpark.
- Formulated data governance structures, guaranteeing data accuracy across 35 data streams.
- Mentored a team of 20 architects and engineers, elevating architecture maturity scores from 70 to 95 within a year.
- Delivered architecture blueprints that streamlined alignment across 7 cross-functional teams in an agile environment.
- Orchestrated architecture working groups, ensuring consistent patterns across all applications and infrastructure pods.
- Pioneered AI-assisted development guardrails integrating with JIRA and maintaining quality assurance standards.

Enterprise Architect

February 2021 – June 2024

Augury, Remote

- Designed a predictive maintenance platform using Microsoft Azure and Kafka, processing 5 billion IoT events annually.
- Enhanced fault detection accuracy by implementing machine learning models in Python and leveraging TensorFlow.
- Consolidated data pipelines into a unified architecture, cutting latency from 3 seconds to milliseconds using Apache Kafka.

- Created a layered data lake with Azure Data Lake and Databricks, enabling instant analytics for 100 industrial clients.
- Deployed security protocols, reducing unauthorized access attempts by 3,000 instances monthly.
- Spearheaded the transition to Dockerized containers, automating deployments across six interconnected service regions.
- Directed a data warehouse redesign on Snowflake, scaling storage from 10 TB to 100 TB and optimizing query speed.
- Implemented incident management strategies, enhancing response times for Level One incidents from one hour to 10 minutes.
- Engineered the architecture for a real-time monitoring system, supporting 500 concurrent engineers across the globe.
- Improved code quality and maintainability through rigorous peer reviews and <GitLab CI/CD pipelines.

Senior Solutions Architect

February 2018 – January 2021

Lob, Remote

- Integrated RESTful API infrastructure, achieving address verification accuracy for 150 million records annually.
- Established a scalable data model using PostgreSQL to optimize processing of 200,000 postal addresses per hour.
- Developed a release management framework that facilitated seamless deployment of production-ready code across all environments.
- Initiated high-availability systems using AWS EC2 and Load Balancing services, ensuring uptime.
- Unified messaging queues via Apache ActiveMQ, supporting a steady workload of 500,000 transactions daily.
- Modernized legacy systems, cutting maintenance costs by implementing an automated testing suite with Selenium and JUnit.

Solutions Architect

June 2016 – January 2018

QuillBot, Remote

- Configured AI-based language processing pipelines using OpenAI's GPT technology, serving 1 million end-users monthly.
- Crafted data-driven applications, processing 10,000 pages of content daily using NLP frameworks like spaCy and NLTK.
- Deployed robust monitoring solutions with Prometheus and Grafana, achieving less than 2 seconds latency for all services.
- Optimized server resources, reducing response times from 8 seconds to under 1 second in high-traffic scenarios.

EDUCATION

University of Texas at Austin, Austin, Texas

May 2014

Bachelor of Science in Computer Science

Advanced Algorithms • Computer Networks • Software Engineering

University of Texas at Austin, Austin, Texas

April 2016

Master of Science in Computer Science

Machine Learning • Distributed Systems • Database Systems

SKILLS & OTHER

Architecture & Design:	Data model design, Solution architecture design, IT security architecture, Data Architecture Design
Integration & APIs:	Implementing APIs, Integration patterns, API versioning, System boundaries
Risk & Security:	Risk mitigation strategies, Threat modeling, Compliance considerations, Security architecture guardrails
Continuous Delivery & DevOps:	CI/CD processes, Release management, Cloud deployment, Scalability
Data Strategies & Management:	Data migrations, Data model evolution, Data governance, Audit and retention
Stakeholder & Mentoring:	Stakeholder management, Mentoring, Decision making, Technical communication